

# Notes: Ardia, Guidotti, and Kroencke (JFE, 2024)

## Efficient Estimation of Bid–Ask Spreads from Open, High, Low, and Close Prices

### Contents

|          |                                                                                             |           |
|----------|---------------------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Big picture</b>                                                                          | <b>2</b>  |
| <b>2</b> | <b>Data, measurement, and key variables</b>                                                 | <b>2</b>  |
| 2.1      | Effective spread . . . . .                                                                  | 2         |
| 2.2      | High-frequency benchmark . . . . .                                                          | 2         |
| 2.3      | Low-frequency inputs . . . . .                                                              | 3         |
| 2.4      | Discrete trading indicator . . . . .                                                        | 3         |
| 2.5      | Data samples . . . . .                                                                      | 3         |
| <b>3</b> | <b>Models and estimators</b>                                                                | <b>3</b>  |
| 3.1      | Log-price model . . . . .                                                                   | 3         |
| 3.2      | Roll close-price estimator . . . . .                                                        | 4         |
| 3.3      | Abdi–Rinaldo close-high-low estimator . . . . .                                             | 4         |
| 3.4      | Open-based discrete estimator . . . . .                                                     | 4         |
| 3.5      | Discrete generalized estimators . . . . .                                                   | 4         |
| 3.6      | Efficient DGE / EDGE . . . . .                                                              | 5         |
| 3.7      | Non-negativity transformation . . . . .                                                     | 5         |
| <b>4</b> | <b>Empirical design and credibility</b>                                                     | <b>5</b>  |
| 4.1      | Simulation design . . . . .                                                                 | 5         |
| 4.2      | Empirical comparison to TAQ benchmark . . . . .                                             | 5         |
| 4.3      | What makes the evidence persuasive . . . . .                                                | 6         |
| 4.4      | Limits . . . . .                                                                            | 6         |
| <b>5</b> | <b>Tables and figures</b>                                                                   | <b>6</b>  |
| 5.1      | Figure 1 + Table 1: why discrete trading matters and how the estimators are built . . . . . | 6         |
| 5.2      | Figure 2 + Figure 3: simulation bias and variance . . . . .                                 | 7         |
| 5.3      | Table 2 + Table 3: simulation results and stock-month summary statistics . . . . .          | 8         |
| 5.4      | Figure 4 + Figure 5 + Table 4: correlations with the high-frequency benchmark . . . . .     | 9         |
| 5.5      | Figure 6 + Figure 7: U.S. stock market applications . . . . .                               | 10        |
| 5.6      | Figure 8: cryptocurrency application . . . . .                                              | 11        |
| <b>6</b> | <b>Short summary</b>                                                                        | <b>12</b> |
| <b>7</b> | <b>Conclusion</b>                                                                           | <b>12</b> |
| 7.1      | Core conclusion . . . . .                                                                   | 12        |
| 7.2      | Economic intuition . . . . .                                                                | 13        |
| 7.3      | Takeaway . . . . .                                                                          | 13        |

# 1 Big picture

- **Question.** How can researchers estimate effective bid–ask spreads accurately when only low-frequency transaction prices are available?
- **Problem with existing estimators.** Popular low-frequency estimators, especially Corwin–Schultz and Abdi–Ranaldo, implicitly assume that prices are observed continuously. In real markets, trades are discrete and sometimes infrequent.
- **Main bias.** When trading is infrequent, daily open, high, low, and close prices often coincide. This makes continuous-observation formulas understate spreads precisely when transaction costs are likely to be high.
- **Main innovation.** The paper derives discrete-time generalized estimators from open, high, low, and close prices. The estimators include analytical correction terms for the probability that the open or close equals the high or low.
- **Efficient estimator.** The authors optimally combine the discrete estimators using a GMM-style weighting scheme to obtain the Efficient Discrete Generalized Estimator, called EDGE.
- **Simulation result.** EDGE remains approximately unbiased even when there is only about one trade per day. Existing continuous-observation estimators shrink toward zero as trading becomes infrequent.
- **Precision result.** When all estimators are unbiased, EDGE has the smallest or near-smallest variance because it chooses which OHLC covariance moments are informative under the spread-volatility regime.
- **Empirical result.** In CRSP daily stock data matched to TAQ effective spreads from 1993–2021, EDGE has the highest correlations with the high-frequency benchmark and lower forecast errors than competing low-frequency estimators.
- **Applications.** The estimator works for historical U.S. stocks since 1926, intraday U.S. stock data, and cryptocurrency pairs where quotes are unavailable or unreliable.
- **Economic takeaway.** Measurement of transaction costs is not a technical detail. Bias in spread estimators can create non-standard errors in empirical finance, especially for small stocks, historical samples, and markets with sparse trading.

## 2 Data, measurement, and key variables

### 2.1 Effective spread

For a trade with transaction price  $P$ , unobserved fundamental price  $\tilde{P}$ , and trade direction  $D \in \{+1, -1\}$ , the relative effective spread is

$$S = \frac{2D(P - \tilde{P})}{\tilde{P}}. \quad (1)$$

**Interpretation:** The effective spread measures how far the transaction price is from the efficient price. High-frequency work tries to proxy  $\tilde{P}$  with quotes. Low-frequency work instead imposes assumptions on price dynamics to infer  $S$  from transaction prices only.

### 2.2 High-frequency benchmark

- The high-frequency benchmark is the effective spread computed from trades matched with quotes using TAQ data.
- The empirical benchmark is based on the Holden–Jacobsen method, denoted HJ.
- The benchmark covers U.S. stocks from 1993–2021 in the CRSP–TAQ merged sample.

**Interpretation:** HJ is not the estimator being proposed. It is the empirical target used to evaluate how well daily-price estimators recover a more direct trade-and-quote measure of transaction costs.

## 2.3 Low-frequency inputs

The low-frequency estimators use only transaction prices:

- $o_t$ : log open price;
- $h_t$ : log high price;
- $l_t$ : log low price;
- $c_t$ : log close price;
- $\eta_t = (h_t + l_t)/2$ : midpoint of the high and low log prices.

**Interpretation:** The key practical appeal is availability. OHLC prices are available for long historical periods, international markets, non-equity assets, and crypto markets where quote data may be costly or unavailable.

## 2.4 Discrete trading indicator

The paper defines

$$\tau_t = \begin{cases} 0, & h_t = l_t = c_{t-1}, \\ 1, & \text{otherwise.} \end{cases} \quad (2)$$

**Interpretation:**  $\tau_t = 0$  captures a day with no useful price movement: either no trade occurs or all observed OHLC values equal the previous close. This is exactly the case ignored by continuous-observation estimators.

## 2.5 Data samples

- **Simulation sample:** 10,000 simulated months; 21 trading days per month; 390 minutes per trading day.
- **CRSP–TAQ matched sample:** U.S. stocks from May 1993 to December 2021 with both daily prices and high-frequency TAQ benchmark spreads.
- **Historical CRSP sample:** U.S. stocks from 1926 to 2021 using daily prices. Open prices are missing from July 1962 through June 1992, so the authors use the close-high-low estimator in that period.
- **High-frequency U.S. stock application:** minute-level prices from Daily TAQ, October 2003 to December 2021.
- **Crypto application:** Binance cryptocurrency pairs from July 2017 to December 2021 using daily, hourly, and minute OHLC prices.

**Interpretation:** The empirical design deliberately spans settings with very different trading intensity and data quality. This is important because the proposed estimator is meant to be broadly applicable.

# 3 Models and estimators

## 3.1 Log-price model

The starting point is the low-frequency log-price representation

$$p_t = \tilde{p}_t + Z_t, \quad (3)$$

where  $\tilde{p}_t$  is the fundamental log price and  $Z_t$  is the bid–ask bounce.

The maintained assumptions are:

- fundamental returns are not serially correlated;
- fundamental returns are uncorrelated with bid–ask bounces;
- bid–ask bounces are uncorrelated and have zero mean.

**Interpretation:** These assumptions are broad enough to include classic low-frequency spread estimators, but the paper relaxes their hidden continuous-trading implication.

### 3.2 Roll close-price estimator

The Roll estimator uses only closing prices:

$$S^2 = -4 \text{Cov}(\Delta c_t, \Delta c_{t-1}). \quad (4)$$

**Interpretation:** Roll is elegant but noisy. In small samples, the covariance can have the wrong sign, generating negative squared spread estimates.

### 3.3 Abdi–Rinaldo close-high-low estimator

The Abdi–Rinaldo estimator uses the previous close and high-low midpoint:

$$S^2 = 4E[(c_{t-1} - \eta_{t-1})(c_{t-1} - \eta_t)]. \quad (5)$$

**Interpretation:** This reduces variance relative to Roll, but still relies on continuous-observation logic. The new method shows it is a special case of a discrete estimator when the correction probability collapses to a constant.

### 3.4 Open-based discrete estimator

The paper derives an estimator from the covariance between open-to-mid and close-to-open returns:

$$S_o^2 = \frac{-8E[(\eta_t - o_t)(o_t - c_{t-1})]}{P[o_t \neq h_t, \tau_t = 1] + P[o_t \neq l_t, \tau_t = 1]}. \quad (6)$$

**Interpretation:** The denominator is the discrete-time correction. If the open frequently equals the high or low, the continuous formula is biased. The correction explicitly accounts for that probability.

### 3.5 Discrete generalized estimators

The paper constructs four building-block estimators:

- OHL: open-high-low based estimator;
- OHLC: open-high-low-close based estimator;
- CHL: close-high-low based estimator;
- CHLO: close-high-low-open based estimator.

**Interpretation:** OHL and OHLC measure spreads at the open; CHL and CHLO measure spreads at the close. Combining them uses the full OHLC information set.

### 3.6 Efficient DGE / EDGE

The two combined moment conditions can be written as  $E[S^2 - x_{i,t}] = 0$ . The efficient estimator solves the one-dimensional GMM problem

$$S_{GMM}^2 = \arg \min_{S^2} \sum_{ij} (S^2 - \mu_i) W_{ij} (S^2 - \mu_j), \quad (7)$$

where  $\mu_i = E[x_{i,t}]$  and  $W = \Omega^{-1}$  is the inverse covariance matrix of the moments.

Because the problem is quadratic, the solution is a weighted average:

$$S_{EDGE}^2 = w_1 E[x_{1,t}] + w_2 E[x_{2,t}], \quad (8)$$

with weights inversely related to the variances of the moment conditions.

**Interpretation:** EDGE gives more weight to the estimator that is more precise in the relevant spread-volatility region. It uses OHLC data efficiently rather than choosing one fixed covariance moment.

### 3.7 Non-negativity transformation

For comparability with prior work, the paper reports nonnegative estimates as

$$\hat{S} = \sqrt{\max\{0, \hat{S}^2\}}. \quad (9)$$

**Interpretation:** This avoids reporting economically meaningless negative spreads. But it can create upward small-sample bias when many negative estimates are reset to zero, so minimizing variance is important.

## 4 Empirical design and credibility

### 4.1 Simulation design

- The simulated fundamental price follows a geometric random walk.
- Daily volatility is set to 3%.
- The baseline spread in the bias simulation is 1%.
- The expected number of trades per day ranges from one to 390.
- Daily OHLC prices are constructed from observed transaction prices; if no trade occurs, the prior close fills the OHLC values.

**Interpretation:** The simulation isolates the exact issue the paper studies: when the same underlying spread is observed through fewer trades, continuous-observation estimators become biased.

### 4.2 Empirical comparison to TAQ benchmark

- Daily-price estimators are compared with HJ effective spreads from TAQ.
- Performance metrics include Pearson correlation, Spearman rank correlation, mean absolute percentage error, root mean squared error, and the fraction of non-positive estimates.
- The paper evaluates performance by exchange, time period, size group, spread group, and trading frequency.

**Interpretation:** The main empirical question is not whether any daily estimator perfectly matches TAQ. It is which low-frequency estimator is most robust across data environments.

### 4.3 What makes the evidence persuasive

- The analytical derivation explains why bias occurs.
- The simulation confirms that bias rises with infrequent trading.
- The TAQ benchmark confirms that the simulation mechanism appears in real stock data.
- The historical and crypto applications show that sampling frequency and market structure materially affect spread estimates.

**Interpretation:** The paper is mainly a measurement-method paper. Its credibility comes from aligning theory, Monte Carlo evidence, and empirical benchmarking.

### 4.4 Limits

- EDGE estimates effective spreads under a covariance-based model; it is not a substitute for high-quality quotes when quotes are reliable and available.
- The estimator still inherits the non-negativity transformation issue in finite samples.
- It does not use all possible information such as volume, order book depth, or Bayesian priors.
- The benchmark depends on trade-quote matching quality, which is itself imperfect.

**Interpretation:** The contribution is not that OHLC data dominate quotes. The contribution is that OHLC data can be used much more efficiently and with less bias than previous low-frequency methods.

## 5 Tables and figures

The visuals below are directly cropped from the original paper and grouped to save space.

### 5.1 Figure 1 + Table 1: why discrete trading matters and how the estimators are built

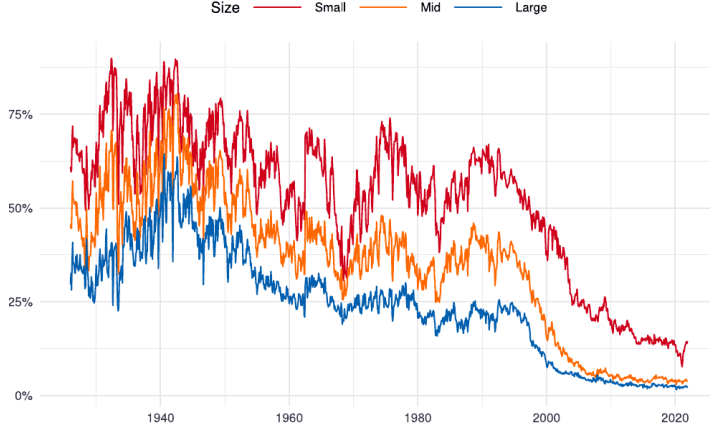
**Read: Figure 1:**

- The probability that the open or close equals the high or low is high historically, especially for small stocks.
- For small stocks, the probability is often above 50% before 2000 and can exceed 75% in earlier decades.
- The probability declines over time as trading becomes more frequent and electronic markets deepen.
- Large stocks have lower coincidence probabilities, consistent with more active trading.

**Read: Table 1:**

- The table lists the four discrete generalized estimators: OHL, OHLC, CHL, and CHLO.
- Each estimator has a correction coefficient  $\pi_o$  or  $\pi_c$  that depends on whether open or close prices equal high or low prices.
- The estimators differ by which OHLC covariance moment they use.

**Economic message:** Discrete observation is not a minor nuisance. The event that open or close prices equal high or low prices is common, so the correction term is central to unbiased spread estimation.



(a) Probability that open/close prices equal high/low prices

**Table 1**  
Discrete generalized estimators.

| Estimators   |                                                                                                                           |                                                                              |      |
|--------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|------|
| OHL          | $S_o^2 = \pi_o \mathbb{E}[(\bar{\eta}_t - \bar{\sigma}_t)(\sigma_t - \eta_{t-1})]$                                        | $S_c^2 = \pi_c \mathbb{E}[(\bar{\eta}_t - c_{t-1})(c_{t-1} - \eta_{t-1})]$   | CHL  |
| OHLC         | $S_o^2 = \pi_o \mathbb{E}[(\bar{\eta}_t - \bar{\sigma}_t)(\sigma_t - c_{t-1})]$                                           | $S_c^2 = \pi_c \mathbb{E}[(\bar{\sigma}_t - c_{t-1})(c_{t-1} - \eta_{t-1})]$ | CHLO |
| Coefficients |                                                                                                                           |                                                                              |      |
| $\pi_o$      | $\pi_o = -8 / \left( \mathbb{P}[\sigma_t \neq h_t, \tau_t = 1] + \mathbb{P}[\sigma_t \neq l_t, \tau_t = 1] \right)$       |                                                                              |      |
| $\pi_c$      | $\pi_c = -8 / \left( \mathbb{P}[c_{t-1} \neq h_{t-1}, \tau_t = 1] + \mathbb{P}[c_{t-1} \neq l_{t-1}, \tau_t = 1] \right)$ |                                                                              |      |
| Prices       |                                                                                                                           |                                                                              |      |
| $o, h, l, c$ | Open, High, Low, Close log-prices.                                                                                        |                                                                              |      |
| $\eta$       | Mid-prices computed as $\eta = (l + h)/2$ .                                                                               |                                                                              |      |

This table reports bid-ask spread estimators obtained from several combinations of open, high, low, and close prices as described in Section 3. The OHL and OHLC estimators measure the spread at the open. The CHL and CHLO estimators measure the spread at the close. The indicator variable  $\tau_t$  is defined in Eq. (7) and the de-meaned

(b) Discrete generalized estimators

## 5.2 Figure 2 + Figure 3: simulation bias and variance

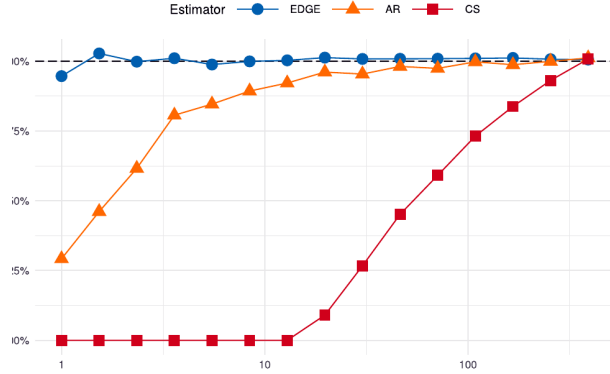
**Read: Figure 2:**

- The true simulated spread is 1%.
- EDGE stays close to the true spread even when the expected number of trades per day is near one.
- CS collapses toward zero at low trading frequencies.
- AR is less biased than CS but still understates the spread when trading is sparse.

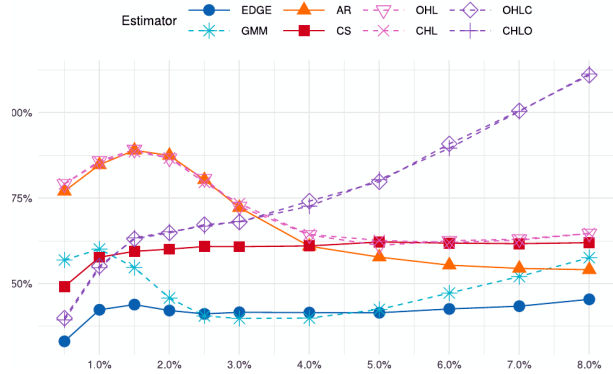
**Read: Figure 3:**

- EDGE has the lowest standard deviation across almost all simulated spread levels.
- CS is precise for small spreads but biased under infrequent trading.
- AR performs better for larger spreads than for very small spreads.
- The GMM combination is useful, but the optimal EDGE weighting improves precision further.

**Economic message:** EDGE dominates because it tackles both problems: it corrects infrequent-trading bias and reduces variance by choosing the best information from OHLC prices.



(a) Bias of simulated spread estimates



(b) Variance of simulated spread estimates

### 5.3 Table 2 + Table 3: simulation results and stock-month summary statistics

Read: Table 2:

- Under frequent trading, most estimators are close to the true spreads, but EDGE has low standard deviations.
- Under infrequent trading, CS severely understates spreads: for a true 1% spread it reports about 0.04%.
- EDGE remains close to the true value in both frequent and infrequent trading panels.
- ROLL is noisy and often has high dispersion.

Read: Table 3:

- The empirical sample contains 1,637,621 stock-month observations.
- EDGE has mean spread 2.11%, median 1.00%, and standard deviation 3.37%.
- EDGE has the highest Pearson correlation with HJ at 78.86% and the highest Spearman correlation at 66.68%.
- EDGE also has the lowest MAPE, RMSE, and fraction of non-positive estimates among the daily-price estimators.

**Economic message:** The simulation and benchmark table tell the same story: the estimator that corrects discrete trading and combines moments efficiently is both less biased and more accurate.

Table 2  
Monthly estimates from simulated daily prices.

|                             |      | EDGE   | OHLc   | CHLo   | OHL    | CHL    | AR     | CS     | ROLL   |
|-----------------------------|------|--------|--------|--------|--------|--------|--------|--------|--------|
| Panel A: Frequent Trading   |      |        |        |        |        |        |        |        |        |
| $S = 0.50\%$                | Mean | 0.44   | 0.46   | 0.46   | 0.79   | 0.79   | 0.70   | 0.60   | 1.44   |
|                             | (sd) | (0.33) | (0.40) | (0.39) | (0.79) | (0.79) | (0.77) | (0.49) | (1.43) |
| $S = 1.00\%$                | Mean | 0.90   | 0.88   | 0.88   | 1.03   | 1.03   | 0.95   | 1.03   | 1.59   |
|                             | (sd) | (0.42) | (0.55) | (0.55) | (0.86) | (0.86) | (0.85) | (0.58) | (1.49) |
| $S = 3.00\%$                | Mean | 2.88   | 2.87   | 2.88   | 2.92   | 2.93   | 2.92   | 2.93   | 2.95   |
|                             | (sd) | (0.41) | (0.69) | (0.69) | (0.73) | (0.72) | (0.70) | (0.61) | (1.83) |
| $S = 5.00\%$                | Mean | 4.87   | 4.86   | 4.87   | 4.92   | 4.93   | 4.97   | 4.90   | 4.90   |
|                             | (sd) | (0.42) | (0.81) | (0.81) | (0.62) | (0.62) | (0.58) | (0.61) | (2.14) |
| $S = 8.00\%$                | Mean | 7.84   | 7.78   | 7.79   | 7.88   | 7.89   | 7.99   | 7.86   | 7.93   |
|                             | (sd) | (0.45) | (1.11) | (1.10) | (0.64) | (0.64) | (0.54) | (0.62) | (2.63) |
| Panel B: Infrequent Trading |      |        |        |        |        |        |        |        |        |
| $S = 0.50\%$                | Mean | 0.71   | 0.77   | 0.79   | 0.89   | 0.91   | 0.65   | 0.02   | 1.44   |
|                             | (sd) | (0.75) | (0.87) | (0.88) | (0.96) | (0.97) | (0.73) | (0.07) | (1.42) |
| $S = 1.00\%$                | Mean | 0.95   | 0.99   | 0.99   | 1.11   | 1.10   | 0.81   | 0.04   | 1.56   |
|                             | (sd) | (0.83) | (0.97) | (0.96) | (1.03) | (1.04) | (0.80) | (0.10) | (1.47) |
| $S = 3.00\%$                | Mean | 2.89   | 2.76   | 2.76   | 2.86   | 2.86   | 2.26   | 0.35   | 2.89   |
|                             | (sd) | (0.83) | (1.23) | (1.23) | (1.20) | (1.19) | (0.92) | (0.36) | (1.82) |
| $S = 5.00\%$                | Mean | 5.02   | 4.89   | 4.92   | 5.01   | 5.04   | 4.04   | 1.17   | 4.83   |
|                             | (sd) | (0.81) | (1.32) | (1.33) | (1.13) | (1.13) | (0.85) | (0.62) | (2.12) |
| $S = 8.00\%$                | Mean | 8.19   | 8.10   | 8.06   | 8.23   | 8.20   | 6.59   | 2.66   | 7.71   |
|                             | (sd) | (0.96) | (1.59) | (1.62) | (1.24) | (1.26) | (0.94) | (0.96) | (2.65) |

(a) Simulation means and standard deviations

| Table 3<br>Summary statistics |           |             |            |           |                         |                         |             |             |             |
|-------------------------------|-----------|-------------|------------|-----------|-------------------------|-------------------------|-------------|-------------|-------------|
|                               | N<br>(1)  | Mean<br>(%) | Med<br>(%) | Sd<br>(%) | Cor <sub>p</sub><br>(%) | Cor <sub>s</sub><br>(%) | MAPE<br>(%) | RMSE<br>(1) | FNPE<br>(%) |
| EDGE                          | 1,637,621 | 2.11        | 1.00       | 3.37      | 78.86                   | 66.68                   | 16.21       | 1.23        | 25.63       |
| OHLc                          | 1,637,621 | 2.22        | 1.05       | 3.57      | 69.87                   | 57.17                   | 18.83       | 1.37        | 29.74       |
| CHLo                          | 1,637,621 | 2.01        | 0.85       | 3.56      | 74.26                   | 58.77                   | 17.08       | 1.27        | 30.97       |
| OHL                           | 1,637,621 | 2.35        | 1.21       | 3.65      | 69.95                   | 54.64                   | 20.47       | 1.49        | 29.97       |
| CHL                           | 1,637,621 | 2.16        | 1.03       | 3.67      | 73.83                   | 55.44                   | 18.53       | 1.41        | 31.30       |
| AR                            | 1,637,621 | 1.70        | 0.95       | 2.50      | 68.13                   | 53.55                   | 19.90       | 1.41        | 31.87       |
| CS                            | 1,637,621 | 0.66        | 0.28       | 1.10      | 45.55                   | 33.77                   | 35.90       | 2.61        | 29.18       |
| ROLL                          | 1,637,621 | 2.47        | 1.39       | 4.09      | 55.22                   | 41.38                   | 24.53       | 1.80        | 32.60       |
| HJ                            | 1,637,621 | 1.89        | 0.75       | 2.73      | -                       | -                       | -           | -           | -           |

The table reports summary statistics of stock-month spread estimates from daily prices in the sample period 1993-2021 (CRSP-TAQ merged sample). Negative spread estimates are set to zero, and we drop the stock-month estimate for all the estimators if it is missing for any of them. The table reports the number of stock-months (N), the mean (Mean), median (Med), and standard deviation (Sd) of the estimates, their Pearson's (Cor<sub>p</sub>) and Spearman's (Cor<sub>s</sub>) correlation with the HJ

(b) Empirical summary statistics

## 5.4 Figure 4 + Figure 5 + Table 4: correlations with the high-frequency benchmark

**Read: Figure 4:**

- EDGE has the highest cross-sectional correlation with HJ for most of the 1993–2021 sample.
- All estimators become noisier in recent years as spreads become very small.
- ROLL and CS are less stable than EDGE.

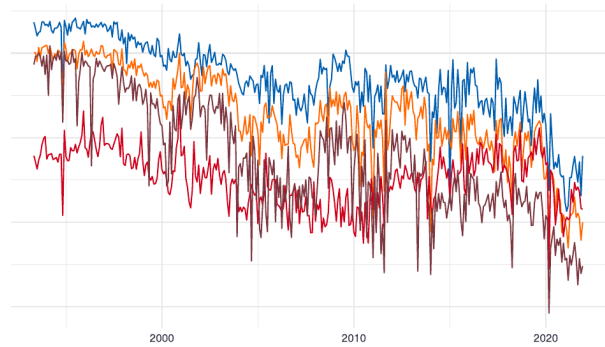
**Read: Figure 5:**

- EDGE has higher time-series correlations with HJ across size deciles.
- The advantage is especially large for smaller stocks where infrequent trading is more relevant.
- CS performs poorly for small size deciles and improves only for the largest stocks.

**Read: Table 4:**

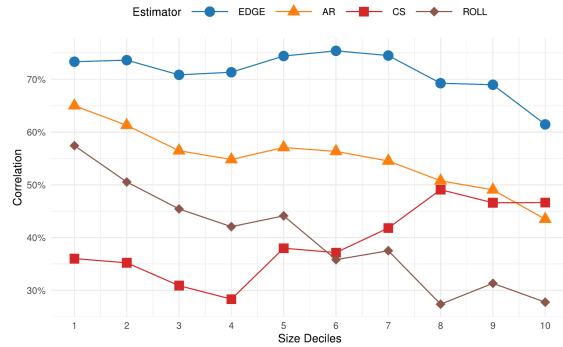
- EDGE is highest across NYSE, AMEX, and NASDAQ.
- EDGE is highest in every time subperiod from 1993–1996 through 2016–2021.
- EDGE is highest across size, spread, and trading-frequency quintiles.

**Economic message:** The estimator's advantage is not confined to one period or one exchange. It is robust across exactly the subsamples where trading intensity and spread size differ.



correlation with the HJ benchmark.  
cross-sectional Pearson's correlations between stock-month spread estimates from daily prices and the HJ benchmark for each month in the sample

(a) Cross-sectional correlation with HJ



time-series correlation with the HJ benchmark.

(b) Time-series correlation with HJ

| Table 4<br>Pearson's correlation with the HJ benchmark. |              |       |       |       |       |       |       |       |
|---------------------------------------------------------|--------------|-------|-------|-------|-------|-------|-------|-------|
|                                                         | EDGE         | OHLC  | CHLO  | OHL   | CHL   | AR    | CS    | ROLL  |
| Panel A: Analysis by Market Exchange                    |              |       |       |       |       |       |       |       |
| NYSE                                                    | <b>64.94</b> | 53.22 | 61.84 | 52.15 | 58.43 | 46.79 | 45.87 | 29.59 |
| AMEX                                                    | <b>68.99</b> | 57.77 | 67.84 | 59.26 | 68.23 | 61.05 | 38.32 | 48.15 |
| NASDAQ                                                  | <b>78.16</b> | 68.62 | 73.24 | 68.87 | 72.99 | 67.03 | 41.09 | 54.81 |
| Panel B: Analysis by Time Period                        |              |       |       |       |       |       |       |       |
| 1993-1996                                               | <b>82.93</b> | 75.94 | 76.83 | 77.68 | 78.57 | 75.31 | 46.94 | 70.23 |
| 1997-2000                                               | <b>78.47</b> | 68.24 | 73.40 | 68.95 | 73.41 | 69.20 | 45.06 | 60.19 |
| 2001-2002                                               | <b>73.04</b> | 60.47 | 70.32 | 61.44 | 69.59 | 67.31 | 40.70 | 59.00 |
| 2003-2007                                               | <b>67.65</b> | 57.56 | 63.45 | 57.26 | 61.97 | 57.34 | 33.87 | 38.10 |
| 2008-2011                                               | <b>69.89</b> | 62.16 | 64.00 | 61.49 | 62.26 | 59.17 | 33.99 | 43.70 |
| 2012-2015                                               | <b>60.78</b> | 51.41 | 55.93 | 52.09 | 55.64 | 53.14 | 37.29 | 29.24 |
| 2016-2021                                               | <b>53.98</b> | 46.72 | 43.97 | 46.48 | 43.11 | 40.78 | 39.34 | 22.11 |
| Panel C: Analysis by Market Capitalization              |              |       |       |       |       |       |       |       |
| Size quintile 1                                         | <b>74.35</b> | 63.60 | 69.90 | 64.86 | 70.44 | 65.00 | 37.16 | 56.08 |
| Size quintile 2                                         | <b>71.29</b> | 60.36 | 66.68 | 60.39 | 66.33 | 56.08 | 30.20 | 44.31 |
| Size quintile 3                                         | <b>75.13</b> | 65.09 | 70.32 | 63.12 | 67.28 | 57.22 | 38.41 | 40.11 |
| Size quintile 4                                         | <b>72.55</b> | 62.93 | 68.07 | 59.63 | 63.44 | 53.02 | 44.60 | 32.90 |
| Size quintile 5                                         | <b>66.65</b> | 57.77 | 61.32 | 54.24 | 56.17 | 47.31 | 47.24 | 30.31 |
| Panel D: Analysis by Spread Size                        |              |       |       |       |       |       |       |       |
| Spread quintile 1                                       | <b>17.84</b> | 15.62 | 16.56 | 15.43 | 15.21 | 14.18 | 12.64 | 9.60  |
| Spread quintile 2                                       | <b>45.66</b> | 39.59 | 41.73 | 34.67 | 34.15 | 30.35 | 32.79 | 15.06 |
| Spread quintile 3                                       | <b>61.98</b> | 52.28 | 57.80 | 48.88 | 52.46 | 44.72 | 40.82 | 24.64 |
| Spread quintile 4                                       | <b>67.76</b> | 55.55 | 64.44 | 55.32 | 63.21 | 55.22 | 37.74 | 38.98 |
| Spread quintile 5                                       | <b>71.38</b> | 60.78 | 66.08 | 62.57 | 67.24 | 61.83 | 33.15 | 55.06 |
| Panel E: Analysis by Trading Frequency                  |              |       |       |       |       |       |       |       |
| Numtrd quintile 1                                       | <b>74.77</b> | 65.88 | 69.12 | 67.68 | 70.37 | 67.81 | 40.02 | 65.10 |
| Numtrd quintile 2                                       | <b>79.15</b> | 69.32 | 74.45 | 69.55 | 74.17 | 69.58 | 51.59 | 52.00 |
| Numtrd quintile 3                                       | <b>75.41</b> | 65.77 | 70.94 | 63.92 | 67.92 | 60.98 | 50.42 | 40.53 |
| Numtrd quintile 4                                       | <b>67.17</b> | 58.53 | 62.78 | 56.00 | 58.71 | 52.54 | 48.98 | 32.41 |
| Numtrd quintile 5                                       | <b>55.48</b> | 48.05 | 50.02 | 45.23 | 45.78 | 39.36 | 43.91 | 22.29 |

Table 4: Pearson correlations with the HJ benchmark

## 5.5 Figure 6 + Figure 7: U.S. stock market applications

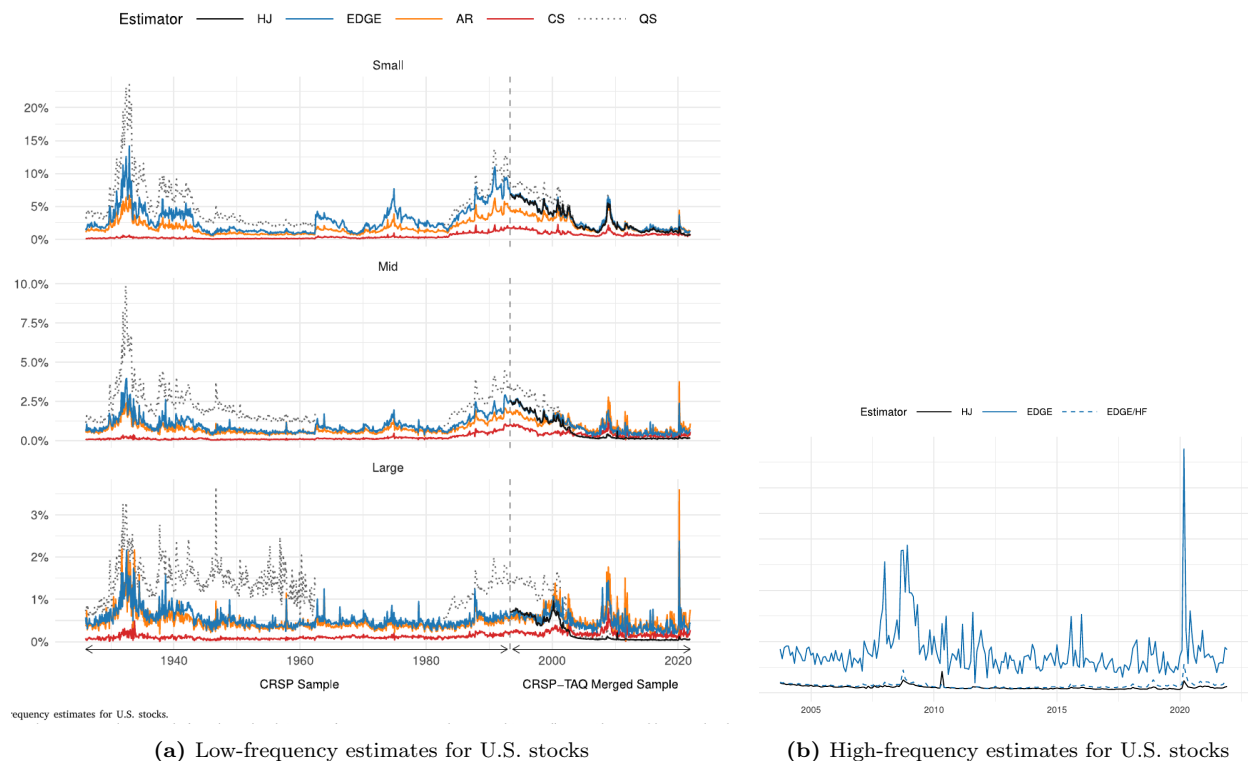
**Read: Figure 6:**

- For small stocks, EDGE produces high spreads during the 1930s, early 1960s, mid-1970s, and early 1990s, matching periods of market stress and high transaction costs.
- CS and AR tend to understate spreads in older periods, especially before TAQ data are available.
- In the post-1993 matched sample, EDGE tracks the HJ benchmark more closely than the alternatives.

**Read: Figure 7:**

- For large-cap stocks, daily EDGE estimates can be noisy and upward biased because spreads are extremely small.
- Using minute-level prices with the same estimator, EDGE/HF, moves estimates much closer to HJ.
- The result shows that the estimator can bridge low- and high-frequency data.

**Economic message:** Daily OHLC prices are valuable for long history, but intraday OHLC prices can dramatically improve precision when spreads are tiny. The same estimator can be used at both frequencies.



## 5.6 Figure 8: cryptocurrency application

### Read:

- The figure compares daily, hourly, and minute estimates for Binance cryptocurrency pairs.
- AR and CS change dramatically with sampling frequency: daily estimates are much higher than hourly estimates, which are much higher than minute estimates.
- EDGE estimates are much less sensitive to the sampling frequency.
- The paper reports that EDGE stays in a narrow average range around 0.68%–0.70%, while AR and CS can differ by roughly an order of magnitude across frequencies.

**Economic message:** A good estimator should not produce radically different transaction-cost estimates merely because the researcher samples daily instead of hourly prices. EDGE greatly reduces this measurement instability.

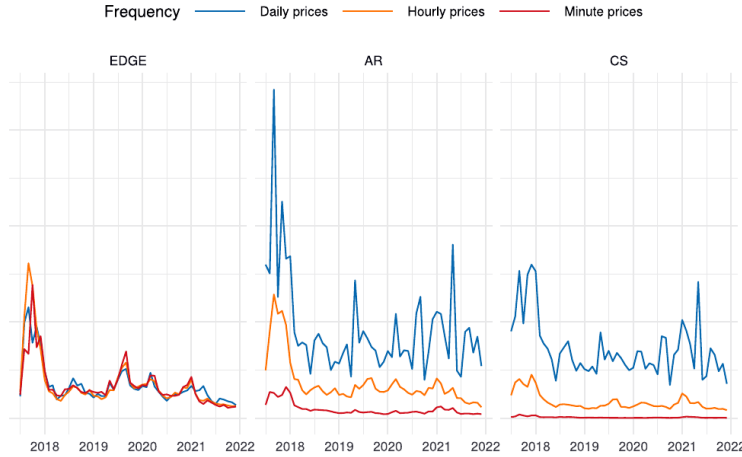


Figure 8: low- and high-frequency estimates for cryptocurrencies

## 6 Short summary

- The paper develops a new estimator of effective bid–ask spreads from OHLC transaction prices.
- Existing low-frequency estimators assume continuous price observation and are biased when trading is infrequent.
- The bias is important empirically because open or close prices often coincide with high or low prices, especially for small stocks and old samples.
- The authors derive discrete generalized estimators that correct this problem analytically.
- They combine the estimators efficiently into EDGE using a GMM-style variance-minimizing weighting scheme.
- Simulations show that EDGE remains unbiased when other estimators collapse toward zero under infrequent trading.
- In CRSP–TAQ data, EDGE has the highest correlations and lowest errors relative to the high-frequency benchmark.
- Historical estimates show that older small-stock spreads are much larger than continuous-observation estimators imply.
- Intraday OHLC data reduce variance and make estimates closer to the benchmark when spreads are tiny.
- Crypto results show that EDGE is less sensitive to the arbitrary choice of daily, hourly, or minute sampling frequency.

## 7 Conclusion

### 7.1 Core conclusion

Ardia, Guidotti, and Kroenke show that low-frequency bid–ask spread estimation needs to account for the fact that transaction prices are observed discretely. Once this is done, OHLC prices contain enough information to produce an estimator that is less biased and more precise than existing daily-price methods.

## **7.2 Economic intuition**

When a stock trades rarely, its high, low, open, and close prices can mechanically coincide. A formula designed for continuously observed prices treats this as evidence of a small spread, even though infrequent trading usually means transaction costs are large. The paper corrects that logic by conditioning on the probability of price coincidences and by choosing the OHLC covariance moments with the lowest variance.

## **7.3 Takeaway**

The main lesson is that measurement design matters. For historical samples, international markets, bonds, crypto, and other settings without reliable quote data, EDGE provides a practical way to measure transaction costs while reducing the bias and sampling-frequency instability of previous estimators.